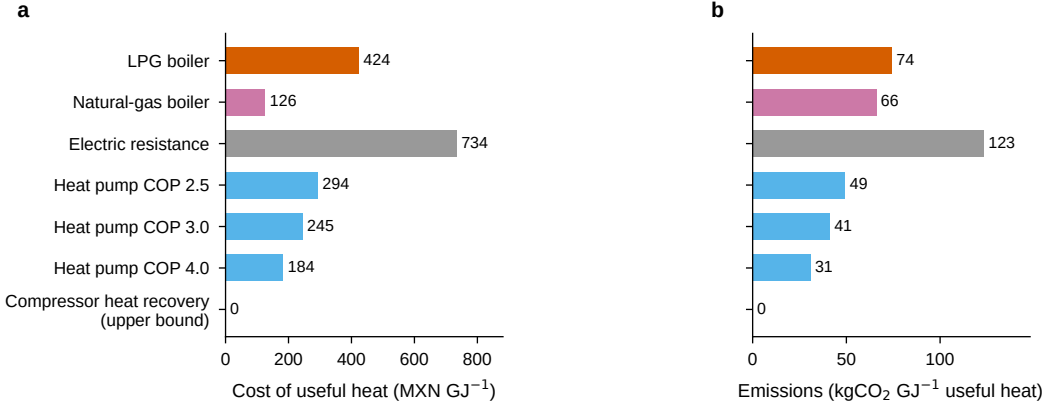
a
LPG boiler
Natural-gas boiler
Electric resistance
Heat pump COP 2.5
Heat pump COP 3.0
Heat pump COP 4.0
Compressor heat recovery (upper bound)
424
126
734
294
245
184
0
0
200
400
600
800
Cost of useful heat (MXN GJ$^{-1}$)
b
74
66
123
49
41
31
0
0
50
100
Emissions (kgCO$_2$ GJ$^{-1}$ useful heat)